



The ECLIPSE Lab of Prof. Dr. Philipp Pelz at FAU Erlangen-Nürnberg offers a position

Ph.D. Student Dose-efficient Data Fusion for Crystalline Mapping and Imaging of Functional Thin Films (m/f/x)

(subject to personal qualifications, remuneration: salary group E 13 TV-L 100%)

starting **November 1st 2026**.

About the Position

We are seeking a highly motivated PhD student to join the **CorMic Graduate School** at Friedrich-Alexander-Universität Erlangen-Nürnberg (FAU) for an exciting doctoral project focused on **dose-efficient data fusion for crystalline mapping and imaging of functional thin films**.

Starting Date: November 1, 2026

Duration: 3 years and 5 months (until March 31, 2030)

Location: Erlangen, Germany

Project Overview

This doctoral project (DP 1) aims to develop cutting-edge **physics-informed data fusion algorithms** that correlate local crystallinity, crystal orientation, and chemical information from 4D-STEM and 4D-STEM-in-SEM experiments. The primary goal is to significantly enhance the signal-to-noise ratio (SNR) of measurements and apply these methods in experimental settings.

Key Research Areas

1. Multi-modal Data Fusion

- Develop algorithms to fuse elastic and inelastic spectroscopic signals in STEM
- Enhance SNR through correlation of multiple measurement channels
- Extract local structure-property information from functional thin films

2. Beam-sensitive Materials Analysis

- Focus on organic thin films that require low-dose imaging
- Implement dose-efficient measurement strategies
- Preserve sample integrity while maximizing information extraction

3. Specific Applications:

- **Inorganic semiconductor layers**
- **Organic absorber layers and electrolyzers**
- **3D correlation studies combining SECM-AFM and 4D-STEM:**

4. Validation and Comparison

- Compare developed approaches with ensemble statistics from spatially averaging X-ray scattering measurements
- Validate methods against established techniques
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Your Profile

Required Qualifications

- **Master's degree** in Physics, Materials Science, Electrical Engineering, or related fields
- Strong background in **electron microscopy** and/or **data analysis**

- Experience with **Python, Pytorch, or similar machine learning frameworks**
- Excellent analytical and problem-solving skills
- Strong written and oral communication skills in English
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Preferred Qualifications

- Experience with **4D-STEM, EELS, or EDS** techniques
- Knowledge of **machine learning** and **signal processing**
- Familiarity with **thin film characterization** methods
- Experience with **scientific computing** and **data fusion algorithms**

What We Offer

Research Environment

- Access to state-of-the-art **electron microscopy facilities** at CENEM
- Integration into the **CorMic Graduate School** network
- Interdisciplinary research environment with experts in materials science, physics, and chemistry

Professional Development

- Structured PhD program with regular progress reviews
- Opportunities to present at international conferences
- Training in advanced microscopy techniques
- Networking with industry partners and research institutions
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Benefits

- Competitive salary according to German public service scale (TV-L E13, 100%)
- Health insurance and social security benefits
- Modern research facilities and equipment

Application Process

Required Documents

1. **Cover letter** explaining your motivation and research interests
2. **Curriculum vitae** with publication list (if applicable)
3. **Academic transcripts** and certificates
4. **Master's thesis** or equivalent research work
5. **Two letters of recommendation** from academic supervisors

Application Deadline

Applications will be reviewed on a rolling basis until the position is filled. Early applications are encouraged.

How to Apply

Please send your application materials as a single PDF file to: **Email:** philipp.pelz@fau.de

Subject: "PhD Application - CorMic DP1 Data Fusion"

About CorMic Graduate School

The **CorMic Graduate School** is a collaborative initiative focused on advancing microscopy techniques for materials characterization. Our research spans from fundamental physics to applied materials science, with a strong emphasis on developing next-generation analytical tools.

About FAU and CENEM

Friedrich-Alexander-Universität Erlangen-Nürnberg (FAU) is one of Germany's leading research universities, consistently ranked among the top institutions for materials science and physics research.

CENEM (Center for Nanoanalysis and Electron Microscopy) provides world-class electron microscopy facilities and expertise, serving as a hub for advanced materials characterization in Germany.

We are committed to diversity and equal opportunity. We welcome applications from candidates of all backgrounds and encourage women and underrepresented groups to apply.

For questions about the position, please contact Prof. Philipp Pelz or the CorMic Graduate School administration.

For informal inquiries, please get in touch with Philipp Pelz (philipp.pelz@fau.de). Applications from underrepresented minorities are particularly welcome. Your application (in English or German) must include a motivation letter and your CV. Applications will be accepted until the position is filled.